

LING YU QIAN A23CS0301 OPERATION SYSTEM SECTION 7

TASK 1 and Task 2

```
Windows PowerShell
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Install the latest PowerShell for new features and improvements! https://aka.ms/PSWindows

PS C:\Users\User> ssh lingyq@192.168.56.10
lingyq@192.168.56.10's password:
Last login: Thu Apr 10 06:05:09 2025 from 192.168.56.1
lingyq@secr2043:~$ rm -rf *
lingyq@secr2043:~$ mkdir share; touch example.txt share/textfile01.txt
lingyq@secr2043:~$ pwd
/home/lingyq
lingyq@secr2043:~$ ls
example.txt  share
lingyq@secr2043:~$ ls share
textfile01.txt
lingyq@secr2043:~$ ls /
bin          home          media         run           sys
bin.usr-is-merged lib          mnt          sbin          usr
boot        lib.usr-is-merged opt          sbin.usr-is-merged var
dev         lib64        proc         snap
etc         lost+found  root         srv
lingyq@secr2043:~$ ls ..
farkhana  lingyq  mahyuddin  student
lingyq@secr2043:~$ ls ../../
bin          home          media         run           sys
bin.usr-is-merged lib          mnt          sbin          usr
boot        lib.usr-is-merged opt          sbin.usr-is-merged var
dev         lib64        proc         snap
etc         lost+found  root         srv
lingyq@secr2043:~$ ls -R /home
/home:
farkhana  lingyq  mahyuddin  student
ls: cannot open directory '/home/farkhana': Permission denied

/home/lingyq:
example.txt  share

/home/lingyq/share:
textfile01.txt
ls: cannot open directory '/home/mahyuddin': Permission denied
ls: cannot open directory '/home/student': Permission denied
lingyq@secr2043:~$ ls -la ~
.          .bashrc      .profile     example.txt
..         .cache       .ssh         share
.bash_history .config      .sudo_as_admin_successful
.bash_logout .local       .wget-hsts

lingyq@secr2043:~$ ls -l
total 4
-rw-r--r-- 1 lingyq lingyq  0 Apr 10 09:22 example.txt
drwxr-xr-x 2 lingyq lingyq 4096 Apr 10 09:22 share
lingyq@secr2043:~$ ls -lh
total 4.0K
-rw-r--r-- 1 lingyq lingyq  0 Apr 10 09:22 example.txt
drwxr-xr-x 2 lingyq lingyq 4.0K Apr 10 09:22 share
lingyq@secr2043:~$ mkdir linux-lab
lingyq@secr2043:~$ ls
example.txt  linux-lab  share
lingyq@secr2043:~$ mkdir workspace
lingyq@secr2043:~$ ls
example.txt  linux-lab  share  workspace
lingyq@secr2043:~$ cd linux-lab/
lingyq@secr2043:~/linux-lab$ pwd
/home/lingyq/linux-lab
lingyq@secr2043:~/linux-lab$ cd
lingyq@secr2043:~$ pwd
/home/lingyq
lingyq@secr2043:~$ ls
example.txt  linux-lab  share  workspace
lingyq@secr2043:~$ rmdir linux-lab
lingyq@secr2043:~$ ls
example.txt  share  workspace
lingyq@secr2043:~$ mkdir linux-lab
lingyq@secr2043:~$ touch file1.txt
lingyq@secr2043:~$ echo "Hello,world" > file1.txt
lingyq@secr2043:~$ cat file1.txt
Hello,world
lingyq@secr2043:~$ echo "Welcome to Operating System lab exercise" > file1.txt
lingyq@secr2043:~$ cat file1.txt
Welcome to Operating System lab exercise
lingyq@secr2043:~$ echo "This is basic Linux command exercise.">>file1.txt
```

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lingyq@secr2043:~$ cat file1.txt
Welcome to Operating System lab exercise
This is basic Linux command exercise.
lingyq@secr2043:~$ ls -l >> filelist.txt
lingyq@secr2043:~$ cat filelist.txt
total 16
-rw-r--r-- 1 lingyq lingyq  0 Apr 10 09:22 example.txt
-rw-r--r-- 1 lingyq lingyq 79 Apr 10 09:36 file1.txt
-rw-r--r-- 1 lingyq lingyq  0 Apr 10 09:44 filelist.txt
drwxr-xr-x 2 lingyq lingyq 4096 Apr 10 09:26 linux-lab
drwxr-xr-x 2 lingyq lingyq 4096 Apr 10 09:22 share
drwxr-xr-x 2 lingyq lingyq 4096 Apr 10 09:25 workspace
lingyq@secr2043:~$ mv file1.txt file2.txt
lingyq@secr2043:~$ cp file2.txt file3.txt
lingyq@secr2043:~$ cat file2.txt file3.txt
Welcome to Operating System lab exercise
This is basic Linux command exercise.
Welcome to Operating System lab exercise
This is basic Linux command exercise.
lingyq@secr2043:~$ cat file2.txt file3.txt > combined.txt
lingyq@secr2043:~$ ls
combined.txt  file2.txt  filelist.txt  share
example.txt  file3.txt  linux-lab     workspace
lingyq@secr2043:~$ cat combined.txt
Welcome to Operating System lab exercise
This is basic Linux command exercise.
Welcome to Operating System lab exercise
This is basic Linux command exercise.

```

TASK 3

```

lingyq@secr2043:~$ ls -l file2.txt
-rw-r--r-- 1 lingyq lingyq 79 Apr 10 09:36 file2.txt
lingyq@secr2043:~$ chmod u-w file2.txt
lingyq@secr2043:~$ ls -l file2.txt
-r--r--r-- 1 lingyq lingyq 79 Apr 10 09:36 file2.txt
lingyq@secr2043:~$ chmod u+w file2.txt
lingyq@secr2043:~$ ls -l file2.txt
-rw-r--r-- 1 lingyq lingyq 79 Apr 10 09:36 file2.txt

```

TASK 4

```
lingyq@secr2043:~$ grep "Welcome" file2.txt
Welcome to Operating System lab exercise
lingyq@secr2043:~$ grep -n
"linux" file2.txt
Usage: grep [OPTION]... PATTERNS [FILE]...
Try 'grep --help' for more information.
Command 'linux' not found, did you mean:
  command 'linex' from snap linex (0.1.6)
See 'snap info <snapname>' for additional versions.
lingyq@secr2043:~$ find . -name file3.txt
./file3.txt
lingyq@secr2043:~$ ./file3.txt
-bash: ./file3.txt: Permission denied
lingyq@secr2043:~$ |
```

```
lingyq@secr2043:~$ rm file3.txt
lingyq@secr2043:~$ cd ~
lingyq@secr2043:~$ rm -r linux-lab
lingyq@secr2043:~$ rm -r share
lingyq@secr2043:~$ rm example.txt
lingyq@secr2043:~$ ls
combined.txt  file2.txt  filelist.txt  workspace
```

```
lingyq@secr2043:~$ rm file3.txt
lingyq@secr2043:~$ cd ~
lingyq@secr2043:~$ rm -r linux-lab
lingyq@secr2043:~$ rm -r share
lingyq@secr2043:~$ rm example.txt
lingyq@secr2043:~$ ls
combined.txt  file2.txt  filelist.txt  workspace
lingyq@secr2043:~$ nano
lingyq@secr2043:~$ nano myfile.txt
lingyq@secr2043:~$ nano hello.txt
lingyq@secr2043:~$ nano sample.txt
lingyq@secr2043:~$ nano sample.txt
lingyq@secr2043:~$ nano sample.txt
lingyq@secr2043:~$ nano sample.txt
lingyq@secr2043:~$ nano sample.txt
lingyq@secr2043:~$ nano practice.txt
```

TASK 5

```
lingyq@secr2043:~$ sudo apt update
[sudo] password for lingyq:
Get:1 http://security.ubuntu.com/ubuntu noble-security InRelease [126 kB]
Hit:2 http://archive.ubuntu.com/ubuntu noble InRelease
Get:3 http://archive.ubuntu.com/ubuntu noble-updates InRelease [126 kB]
Hit:4 http://archive.ubuntu.com/ubuntu noble-backports InRelease
Fetched 252 kB in 2s (132 kB/s)
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
3 packages can be upgraded. Run 'apt list --upgradable' to see them.
lingyq@secr2043:~$ apt list --upgradable
```

```
lingyq@secr2043:~$ sudo apt upgrade
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
Calculating upgrade... Done
The following upgrades have been deferred due to phasing:
  pci.ids python3-software-properties software-properties-common
0 upgraded, 0 newly installed, 0 to remove and 3 not upgraded.
lingyq@secr2043:~$ sudo apt install build-essential
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
build-essential is already the newest version (12.10ubuntu1).
0 upgraded, 0 newly installed, 0 to remove and 3 not upgraded.
```

TASK 6

```
lingyq@secr2043:~$ ls
combined.txt  file2.txt  filelist.txt  hello.txt  practice.txt  sample.txt  workspace
lingyq@secr2043:~$ cd workspace
lingyq@secr2043:~/workspace$ nano hello_world.c
lingyq@secr2043:~/workspace$ nano hello_world.c
lingyq@secr2043:~/workspace$ gcc hello_world.c -o hello_world
lingyq@secr2043:~/workspace$ ./hello_world
Hello, World!
lingyq@secr2043:~/workspace$ cd workspace
-bash: cd: workspace: No such file or directory
lingyq@secr2043:~/workspace$ nano hello_world.cpp
lingyq@secr2043:~/workspace$ g++ hello_world.cpp -o hello_world2
```

```
lingyq@secr2043:~/workspace$ ./hello_world2
Hello World!
lingyq@secr2043:~/workspace$ nano argument.c
lingyq@secr2043:~/workspace$ gcc argument.c -o argument
```

```
lingyq@secr2043:~/workspace$ nano argument.c
lingyq@secr2043:~/workspace$ gcc argument.c -o argument
lingyq@secr2043:~/workspace$ ./argument farkhana 23
Hello, farkhana! You are 23 years old.
```

```
lingyq@secr2043:~/workspace$ nano input_keyboard.c
```

```
lingyq@secr2043:~/workspace$ gcc input_keyboard.c -o keyboard
```

```
lingyq@secr2043:~/workspace$ ./keyboard
Enter your name: LING YU QIAN
Enter your age: 21
Hello, LING YU QIAN! You are 21 years old.
```

```
lingyq@secr2043:~/workspace$ nano multiply.c
lingyq@secr2043:~/workspace$ gcc multiply.c -o multiply
lingyq@secr2043:~/workspace$ ./multiply 3 5
3 * 5 = 15
```

```
lingyq@secr2043:~/workspace$ nano kernel_info.c
lingyq@secr2043:~/workspace$ gcc kernel_info.c -o kernel_info
lingyq@secr2043:~/workspace$ ./kernel_info
System name: Linux
Node name: secr2043
Kernel release: 6.8.0-57-generic
Kernel version: #59-Ubuntu SMP PREEMPT_DYNAMIC Sat Mar 15 17:40:59 UTC 2025
Machine hardware: x86_64
```